

Project Development Phase

Model Performance Test (Power BI Dashboard Analytics)

Date	October 03,2026
Team ID	SWTID-2026-7210
Project Name	Script Controlled ACL - Restrict Record Access Based On Field Value
Maximum Marks	10 Marks Evaluation Tier

Model Performance Testing & Dashboard Implementation

S.No.	Parameter	Screenshot / Values Specification Log Details
1	Data Rendered	• Data Source Type: Live Relational Database Query Log Stream connected via DirectQuery. • Total Records Loaded: 254,812 historical security audit verification data rows. • Core Fields Mapped: Record_ID, Table_Name, Field_Value_Flag, Execution_Script_Status, Triggered_User_Role, Dynamic_Access_Decision (Allow/Deny), Validation_Latency_ms. <i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>
2	Data Preprocessing	• Column Splitting & Transformations: Applied Power Query M-Engine scripts to extract specific JSON payload strings from raw unstructured database system log files. • Missing Values Management: Replaced blank conditional string tags in 'Field_Value_Flag' with 'Unassigned_Context' to maintain dynamic audit trails accuracy. • Data Type Standardization: Coerced Validation_Latency columns into high-precision Integer format (Fixed Decimal) and assigned structural indexes over Rule_IDs. <i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>
3	Utilization of Data Filters	• Slicers & Context Interceptors Added: Interactive Page-Level Slicers deployed for 'Table_Name', 'Triggered_User_Role', and 'Dynamic_Access_Decision'. • Advanced Field Filters: Implemented relative time filtering models displaying security metrics within the last 24-hour evaluation cycles. • Visual Level Security Interception: Configured cross-filtering constraints across visual blocks to trace critical outlier anomalies automatically when access flags match 'Confidential'. <i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>
4	DAX Queries Used	• Formula 1: Total Evaluation Requests Count Total_Requests = COUNT(SecurityAuditLogs[Record_ID])

		● Formula 2: Dynamic Row Interception Rate % Denial_Rate = DIVIDE(CALCULATE(COUNT(SecurityAuditLogs[Record_ID]), SecurityAuditLogs[Dynamic_Access_Decision] = "Deny"), [Total_Requests], 0) ● Formula 3: Maximum Access Violation Processing Delay Threshold Max_Latency_MS = MAX(SecurityAuditLogs[Validation_Latency_ms]) ● Formula 4: Script Validation Latency Class Range Evaluator Performance_Status = IF([Max_Latency_MS] <= 150, "Optimal SLA Speed", "Requires Token Optimization") <i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>
5	Dashboard design	● No of Visualizations / Graphs: 6 Interactive Dashboard Core Blocks ● Visual 1: Matrix Table Grid tracking dynamic rules logic evaluation breakdowns across user role tiers. ● Visual 2: Key Performance Indicator (KPI) Cards mapping out Total Request Counts and Average Evaluation Latencies. ● Visual 3: Clustered Bar Chart visualizing Access Decision decision streams (Allow vs Deny) grouped against distinct database fields. ● Visual 4: Stacked Column Visual analyzing transaction time metrics distributions to catch system bottlenecks. <i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>
6	Report Design	● No of Visualizations / Graphs: 4 Detailed Regulatory Compliance Pages ● Section 1: Executive Overview Summary highlighting core system vulnerabilities. ● Section 2: Detailed Policy Enforcement Drill-Through Sheets isolating script execution execution parameters and field validation variables. ● Section 3: Chronological Log Analysis Line Ribbon Charts exploring peak exception event timestamps. ● Layout Design Profile: Dark Navy Blue Theme alignment guidelines matching corporate DevSecOps tracking structures. <i>[The Screenshots Of The Projects Are Provided Below For Reference]</i>

